

Tutorial 10 – Bitwise Operators

1. Determine the result of the following operations:

a. 11111100 & 00111111	d. 11111111 10101010
b. 11111111 & 10101010	e. 11111100 ^ 00111111
c. 11111100 00111111	f. 11111111 ^ 10101010

2. The value of p in binary format is 11111111. What is the result of the following expressions?

(i) $p \& \sim p$ (ii) $p | \sim p$ (iii) $p \wedge p$

3. There are 3 bitwise operators AND (&), OR (|) and XOR (^). These operators take two operands: first operand, A, is 1001 0110 and second operand, B, is 0001 1010.

- i. Write the result of A & B.
- ii. Write the result of A | B.
- iii. Write the result of A ^ B.

4. Determine the result of the following operations (operands are in binary):

- i. $\sim 1110\ 0110$
- ii. $1110\ 0011 \& 1010\ 1100$
- iii. $1100\ 1010 | 1010\ 1110$
- iv. $0110\ 1101 \wedge 1100\ 0011$

5. Determine the bit patterns for the following operations:

- i. Left shift 2 bits on the bit pattern 10110011
- ii. Right shift 3 bits on the bit pattern 10110011

6.
 - i. If $11001100 \& m$ yields 10001000, determine the bit pattern for m.
 - ii. If $11001100 | n$ yields 11101110, determine the bit pattern for n.
 - iii. If $11001100 \wedge p$ yields 10101010, determine the bit pattern for p.

7. *Left shift an integer by 1 is equivalent to multiplying the value by 2.*

In order to verify the above statement, convert the following decimal numbers to binary, left shift by 1, then convert the results back to decimal. Use 8 bits in your calculations.

- i. 10
- ii. 28
- iii. 56